

Early Favorable PSA Response Prediction in Metastatic Hormone Sensitive Prostate Cancer

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Reviewer Manuscript Overview:

This research is significant for its potential practical implications for improving the health and lives of patients with metastatic hormone-sensitive prostate cancer (mHSPC). The study is highly motivated and builds upon an extensive body of literature that establishes a strong link between early, deep PSA response and better long-term clinical outcomes, including overall survival, in mHSPC patients.

The overall methods and approach are suitable for the prediction model and research purposes (barring additional questions/comments I provided below). I commend in particular the efforts of the authors to obtain and utilize individual patient data (IPD) from each of the four pivotal mHSPC ARPI trials, as these efforts were instrumental in ensuring potential study/model robustness and reliability.

Reviewer Comments and Questions:

Comments by Manuscript Page and Line Number

- Pg 13 15 – Why was baseline PSA fitted with restricted cubic splines with 4 DoF? Please clearly explain and justify the rationale given this particularly specific fit.
- Pg 13 19 – How about Gleason Score? Considerations for controlling given its link to PSA response and longer term clinical outcomes?
- Pg 13 19 – A key covariate/potential confounding factor that comes to mind for me is the temporal differences between the trials. For example, changes in standard of care (in LPC and mPC), advanced/next generation imaging (NGS) to detect mPC, and other changes in recommended clinical care in this disease area between the time say LATITUDE and ARASENS were conducted. Suggest to control for time or year of treatment (or initiation) across patient and trial cohorts or provide rationale for it's not controlled for.
- Pg 13 20 – Why was AIC used to select model variables? Any considerations for Elastic Net, LASSO, or other ML approaches for variable selection to better handle potential overfitting or multicollinearity?
- Pg 14 2 – Why were tertiles chosen, as opposed to say quartiles for example? Either is fine, but would be helpful to provide a justification.
- Pg 16 6 – What's most interesting to me is that the greatest relative gains in the model were for patients in the middle 5-50 ng/mL group, suggest calling this out. Any ideas why this group has biggest relative gain in accuracy for the model?
- Pg 16 20 – Why were 4-year OS probabilities modeled instead of the typical 5-year OS? Fine if due to data limitations or follow-up counts, but please justify in the text if so.
- Figure E6.1/E6.2 Interesting that the survival probability for upper-middle tertiles is higher for patients without docetaxel (72%) than with docetaxel (66%) (E6.1) and in upper tertile (72% and 78%) (E6.2), at least descriptively. Why is this the case, residual confounding of patient characteristics? Maybe helpful to call out in manuscript, beyond the general limitation

(given the relatively large difference).

Major Comments on Overall Methodology

- My biggest question or concern is the lack of justification around why ENZAMET was specifically used for the external validation dataset. Some commonly established challenges and drawbacks of using only one trial/database for external validation (ENZAMET) include the potential for: overfitting risk and reduced generalizability, especially if additional differences between the internal training/external validation samples (e.g., patient characteristics, years of trial, etc.)

I would suggest the use of leave-one-out cross-validation (LOOCV), which would aggregate model fit by using and rotating each of the 4 pivotal trial databases as the external validation arm, which can help reduce the potential for dataset-specific overfitting and bolster model generalizability with more equitable model evaluation across the 4 trial datasets.

- IF it is not feasible for LOOCV or a more generalizable test/training validation to be done, or if there's a rationale that supports the given internal/external validation approach (with one external dataset) vs. LOOCV or other approaches, then please be sure to provide adequate explanation and justification for your current approach of training/testing and why the authors identified this as the best method and approach. Also, please give a strong rationale and justification for why ENZAMET specifically was chosen as external validation cohort, of the four pivotal Phase 3 trials, and update the limitations section as applicable.

Reviewer #2

(Remarks to the Author)

In this manuscript, Dr. Roy et al. described a novel tool developed to identify early favorable PSA response in mHSPC patients who would benefit from early treatment personalization. They used mHSPC patients treated on ADT+ARPI arm of LATITUDE, TITAN, ARANSES trials (training and internal validation cohort), and ENZAMET (as external validation cohort). They concluded that their model satisfactorily predicts early favorable PSA response (≤ 0.2 ng/mL by 6 months) and needs prospective validation.

The reviewer has the following comments:

1. Please describe the randomization process of pooled data into training (60%) and internal validation (40%) cohorts. Is there a reason for 60%:40 % randomization vs 50%: 50% (which may seem logical).
2. The ENZAMET trial had a planned evaluation of participants (including PSA) every 12 weeks, and allowed prior docetaxel, which is different from other pooled trials, which are industry-sponsored and had more PSA data in the first 6 months. Is there a reason why ENZAMET was selected as an external validation cohort? Please comment.
3. Discussion (Page 19). What are the values of PSA of "patients with lower baseline PSA values" and "patients with higher baseline PSA values. Please be more specific.
4. The number of figures is extensive. For instance, Figures E1.1, E1.2, and E4 can be moved to the supplementary figures.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Revisions and comment responses to my questions are suitable and sufficient. Recommend to accept for publication on my end.

Reviewer #2

(Remarks to the Author)

I think that the authors have adequately addressed the comments made by the reviewers in the revised version of the manuscript.

Therefore, I have no further comments.

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Re: NCOMMS-25-38385-T

Manuscript Number: NCOMMS-25-38385-T

Revision Version: R1

Date of Receipt of Editorial decision: July 21, 2025

Date of Submission of Revision: August 29, 2025

Author Response 1st Revision:

On behalf of my co-authors, I would like to thank the reviewers for the time taken to critically appraise this manuscript and for allowing us the opportunity to respond to their comments. We have addressed, item by item, the reviewers' comments, with responses appearing below. Given the concern on validation, we have performed the leave-one-trial-out validation in line with the reviewer suggestion. The model continues to show good discrimination and calibration. Further, we have shortened the abstract to comply with the word count limits. Revisions have been made to the manuscript with changes tracked and highlighted, and we have submitted the revised marked manuscript. The page and line numbers in our responses refer to the marked manuscript.

Reviewer 1

Reviewer Comments:

Comment 1: Why was baseline PSA fitted with restricted cubic splines with 4 DoF? Please clearly explain and justify the rationale given this particularly specific fit.

Authors' response: We tested models without splines and with spline degrees of freedom (DoF) ranging from 3 to 5. The model with 4 DoF achieved the lowest AIC and thus was chosen for this model. We have mentioned in our manuscript that "Variable selection was based on clinical relevance, with final model specification determined by the Akaike Information Criterion" on page no: 5, line no: 115 to 117.

Comment 2: How about Gleason Score? Considerations for controlling given its link to PSA response and longer term clinical outcomes?

Authors' response: We agree with the reviewer that Gleason score does hold some prognostic value in metastatic prostate cancer. However, pre-ARPI Gleason score was not readily available across the included trials. Further, the reported Gleason scores did not account for difference between clinical Gleason scores from biopsy versus that derived from prostatectomy, old versus new grading system, and the differences in reporting pattern across various countries given these trials were multi-national. For example, Canada and UK report synoptic Gleason score while USA reports maximum Gleason score. For these reasons, we elected not to include Gleason score in our model. We have acknowledged this limitation in the revised manuscript (page no: 15, line no: 335).

Comment 3: A key covariate/potential confounding factor that comes to mind for me is the temporal differences between the trials. For example, changes in standard of care (in LPC and mPC), advanced/next generation imaging (NGS) to detect mPC, and other changes in recommended clinical care in this disease area between the time say LATITUDE and ARASENS were conducted. Suggest to control for time or year of treatment (or initiation) across patient and trial cohorts or provide rationale for it's not controlled for.

Authors' response: We appreciate the reviewer's point regarding temporal differences between trials. While an excellent point, fundamentally this will always exist given the time between trial design and reporting of results in mHSPC trials will be 7+ years (1-2 years to design and activate trial, 2-5 years to enroll, 3-5+ years of follow-up). Fortunately, all four included trials have established the standard-of-care (SOC) systemic therapies for mHSPC and collectively define the evidence base guiding current practice, thereby reducing variability in therapeutic context. Importantly, our leave one out cross validation (LOOCV) analysis inherently addresses temporal heterogeneity, as each trial (spanning different eras) served as an independent validation cohort, demonstrating consistent performance across time frames. The LOOCV analysis is described on page no: 6, line no: 135 to 140 and reported on page no: 8, line no: 182 to 184. We have also highlighted in the discussion that while these trials underpin the

present SOC, emerging tools such as PET imaging and evolving treatment paradigms are not fully captured (page no: 15, line no: 328 to 334), and future work will evaluate model performance in these contexts.

Comment 4: Why was AIC used to select model variables? Any considerations for Elastic Net, LASSO, or other ML approaches for variable selection to better handle potential overfitting or multicollinearity?

Authors' response: We thank the reviewer for this suggestion. We used AIC for variable selection because it provides a well-established, information-theoretic framework that balances model fit and complexity, favoring parsimony and reducing the risk of overfitting. This approach is widely applied in clinical prediction model development where interpretability and generalizability are essential. As part of our sensitivity analyses, we evaluated machine learning methods, including Random Forests, which are robust to multicollinearity and capable of modeling complex nonlinear relationships. Random Forest achieved performance similar to our AIC-based logistic regression model in the external validation cohort (AUC: 0.83 [0.79-0.86] vs 0.82 [0.78-0.85]; Brier score: 0.17 vs 0.16), confirming that our simpler logistic regression model achieved comparable accuracy (page no: 11, line no: 243 to 245 and supplementary results R2). Given its superior interpretability and ease of clinical implementation, we prioritized the logistic regression model while acknowledging that more complex approaches did not offer meaningful incremental benefit (page no: 12, line no: 259 to 261).

Comment 5: Why were tertiles chosen, as opposed to say quartiles for example? Either is fine, but would be helpful to provide a justification.

Authors' response: We thank the reviewer for this observation. We selected tertiles to ensure adequate sample sizes and event counts within each subgroup. This avoided over-stratification, which could lead to unstable estimates and wide confidence intervals. We have mentioned this on page no: 6, line no: 122 to 124. To confirm robustness, we performed a sensitivity analysis using quartile-based stratification (page no: 9, line no: 198 to 201; supplementary figure 4.1 and 4.2), which showed similar trends, supporting our choice of tertiles for the primary analysis.

Comment 6: What's most interesting to me is that the greatest relative gains in the model were for patients in the middle 5-50 ng/mL group, suggest calling this out. Any ideas why this group has biggest relative gain in accuracy for the model?

Authors' response: We thank the reviewer for this insightful observation, which led us to re-examine the IPA values across baseline PSA strata. During this process, we identified that the previously reported IPA for the multivariable logistic regression model (NADIR model) in the >50 ng/mL PSA group was not updated from an earlier draft version of the table during manuscript assembly. This was a reporting oversight that did not affect the main aspects of analysis or the primary findings.

The correct IPA for the NADIR model in this group is 10.9, compared to 6.0 for the PSA-only model. With this corrected, the NADIR model demonstrates consistently greater predictive accuracy than the PSA-only model across all three PSA subgroups (≤ 5 ng/mL, >5 – 50 ng/mL, >50 ng/mL), with relatively higher incremental gains in those with PSA >5 ng/mL. We have verified all other subgroup results and confirm their accuracy. The manuscript has been updated accordingly, and we appreciate the reviewer's comment for helping us strengthen the clarity and accuracy of our report.

Comment 7: Why were 4-year OS probabilities modeled instead of the typical 5-year OS? Fine if due to data limitations or follow-up counts, but please justify in the text if so.

Authors' response: We appreciate this comment. As the reviewer highlighted, the follow-up was limited beyond 4 years in the training and validation cohorts. This is why we reported the 4-year overall survival estimates. We have now mentioned this clearly in our statistical consideration section – "Overall survival (OS) was reported at 4 years because follow-up beyond 4 years was limited in the training and validation cohorts. (page no: 7, line no: 145 to 147)". Further, while 5 year OS estimates are preferred in the context of oncology, studies on metastatic prostate cancer have often reported 3-year overall survival estimates.

Comment 8: Interesting that the survival probability for upper-middle tertiles is higher for patients without docetaxel (72%) than with docetaxel (66%) (E6.1) and in upper tertile (72% and 78%) (E6.2), at least descriptively. Why is this the case, residual confounding of patient characteristics? Maybe helpful to call out in manuscript, beyond the general limitation (given the relatively large difference).

Authors' response: We agree with the astute comment from the reviewer. We have described the 4-year OS estimates in the text (page no: 10, line no: 224 to 232). Further, we have discussed this finding in the discussion – "In our study, we did not observe an OS benefit with addition of docetaxel in those with a relatively higher predicted probability of early PSA response to ≤ 0.2 ng/mL by 6 months. Given the post-hoc exploratory nature, potential for residual confounding, and the lack of a trial randomizing to ADT+ARPI +/- docetaxel, this finding is merely hypothesis generating and is liable to both type I and type II errors. Furthermore, patients referred for docetaxel on trial might had other poor risk features not captured in the model which needs further exploration with additional biomarkers (page no: 14, line no: 311 to 317)."

Comment 9: Major Comments on Overall Methodology

My biggest question or concern is the lack of justification around why ENZAMET was specifically used for the external validation dataset. Some commonly established challenges and drawbacks of using only one trial/database for external validation (ENZAMET) include the potential for: overfitting risk and reduced generalizability,

especially if additional differences between the internal training/external validation samples (e.g., patient characteristics, years of trial, etc.)

I would suggest the use of leave-one-out cross-validation (LOOCV), which would aggregate model fit by using and rotating each of the 4 pivotal trial databases as the external validation arm, which can help reduce the potential for dataset-specific overfitting and bolster model generalizability with more equitable model evaluation across the 4 trial datasets.

If it is not feasible for LOOCV or a more generalizable test/training validation to be done, or if there's a rationale that supports the given internal/external validation approach (with one external dataset) vs. LOOCV or other approaches, then please be sure to provide adequate explanation and justification for your current approach of training/testing and why the authors identified this as the best method and approach. Also, please give a strong rationale and justification for why ENZAMET specifically was chosen as external validation cohort, of the four pivotal Phase 3 trials, and update the limitations section as applicable.

Authors' response: We thank the reviewer for highlighting the importance of robust validation. We used patients LATITUDE, TITAN, and ARASENS to randomly split into training and internal validation dataset. External validation employed individual patient data from the ARPI arm of the ENZAMET trial to assess the locked model's generalizability. In line with the reviewer's suggestion and with expanded access to all four phase 3 trials (LATITUDE, TITAN, ARASENS, ENZAMET), we have now performed Leave-One-Trial-Out Validation (LOOCV). In this approach, each trial served as an external validation set while training was conducted on the remaining three trials. The model demonstrated consistent performance across all trials (AUC 0.78–0.82; Brier 0.16–0.19; Supplementary Table 2), supporting its generalizability and addressing concerns of dataset-specific overfitting. In the manuscript we have corresponding changes in the methods section (page no: 6, line no: 135 to 140) and results section (page no: 8, line no: 182 to 184).

Reviewer 2

Reviewer Comments:

In this manuscript, Dr. Roy et al. described a novel tool developed to identify early favorable PSA response in mHSPC patients who would benefit from early treatment personalization. They used mHSPC patients treated on ADT+ARPI arm of LATITUDE, TITAN, ARANSES trials (training and internal validation cohort), and ENZAMET (as external validation cohort). They concluded that their model satisfactorily predicts early favorable PSA response (≤ 0.2 ng/mL by 6 months) and needs prospective validation.

The reviewer has the following comments:

Comment 1: Please describe the randomization process of pooled data into training (60%) and internal validation (40%) cohorts. Is there a reason for 60%:40 % randomization vs 50%: 50% (which may seem logical).

Authors' Response: We thank the reviewer for this question. We used a 60:40 split to prioritize a larger training cohort for model development, ensuring stable coefficient estimation while retaining a sufficiently large internal validation cohort for performance testing. This approach is consistent with prior predictive modeling studies where slightly larger training sets are preferred to optimize parameter estimation, particularly in settings with multiple covariates. We have clarified this in the Methods section and explicitly stated that allocation was performed using random assignment without stratification (page no: 5, line no: 104 to 107).

Comment 2: The ENZAMET trial had a planned evaluation of participants (including PSA) every 12 weeks, and allowed prior docetaxel, which is different from other pooled trials, which are industry-sponsored and had more PSA data in the first 6 months. Is there a reason why ENZAMET was selected as an external validation cohort? Please comment.

Authors' response: We thank the reviewer for this comment. We used patients LATITUDE, TITAN, and ARASENS to randomly split into training and internal validation dataset. Subsequently, external validation employed individual patient data from the ARPI arm of the ENZAMET trial to assess the locked model's generalizability. In line with the reviewer's suggestion, we performed Leave-One-Trial-Out Validation (LOOCV). In this approach, each trial served as an external validation set while training was conducted on the remaining three trials. The model demonstrated consistent performance across all trials (AUC 0.78–0.82; Brier 0.16–0.19; Supplementary Table 2), supporting its generalizability and addressing concerns of dataset-specific overfitting. In the manuscript we have corresponding changes in the methods section (page no: 6, line no: 135 to 140) and results section (page no: 8, line no: 182 to 184).

Comment 3: Discussion (Page 19). What are the values of PSA of “patients with lower baseline PSA values” and “patients with higher baseline PSA values. Please be more specific.

Authors' response: We thank the reviewer for highlighting this. We have now mentioned in the discussion – “In patients with lower baseline PSA levels (≤ 5 ng/mL), the multivariable model performed similarly to, or slightly better than, PSA alone. This likely reflects the inherently higher probability of achieving an early nadir in this subgroup, reducing the clinical need for additional predictive modeling. Conversely, in patients with higher baseline PSA (> 5 ng/mL) - where nadir prediction remains an unmet clinical need - the multivariable model demonstrated superior predictive performance compared to baseline PSA alone”. (page no: 12, line no: 268 and 271)

Comment 4: The number of figures is extensive. For instance, Figures E1.1, E1.2, and E4 can be moved to the supplementary figures.

Authors' response: We appreciate this comment from the reviewer. The supplementary figures E1.1, E1.2 and E4 were part of the supplementary materials. All the tables and figures starting with "E" were part of the supplementary materials. Following the Nature author guidelines, we have now edited them, and they are now simply "Supplementary Figure 1.1", "Supplementary Figure 1.2", "Supplementary Figure 4" etc.